MIL-W-17265/7A(SHIPS)
4 January 1974
SUPERSEDING
MIL-W-17265/7(SHIPS)
1 March 1968
(See 6.2)

MILITARY SPECIFICATION

WINCH, DRUM, POWER OPERATED

(SPANWIRE WINCH, AUTOMATIC TENSION CONTROL,

TYPE 8.5-SW(AT)-1D and 8.5-SW(AT)-1D/6.0-2G)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for type 8.5-SW(AT)-1D and type 8.5-SW(AT)-1D/6.0-2G spanwire winches.

1.2 Classification. Winches shall be of the following types, as specified (see 6.1):

Type 8.5-SW(AT)-1D - Single drum spanwire winch rated 8500 pounds at 300 feet per minute(fpm).

Type 8.5-SW(AT)-1D/6.0-2G - Single drum double gypsy spanwire winch with drum unit rated 8500 pounds at 300 fpm and gypsy unit rated 6,000 pounds at 120 and 60 fpm.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
RR-W-410 - Wire Rope and Strand.

MILITARY
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C., and Associated Switching Devices, Naval Shipboard.
MIL-W-15802 - Winch, Gypsy, Power Operated, Electric.
MIL-B-16392 - Brakes, Magnet, Naval Shipboard.
MIL-M-17060 - Motors, 60 Hertz, Alternating-Current, Integral Horsepower, (Shipboard Use).
MIL-M-17265 - Winch, Drum, Power Operated (Electric-Hydraulic, Variable Speed).
MIL-C-23742 - Control Centers, Alternating Current, Naval Shipboard.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Winch types 8.5-SW(AT)-1D and 8.5-SW(AT)-1D/6.0-2G shall be in accordance with MIL-W-17265, except as specified herein.

3.2 Construction.

3.2.1 Type 8.5-SW(AT)-1D. Type 8.5-SW(AT)-1D shall consist of a single drum single gypsy, spanwire winch mounted on a single bedplate(see table I).

3.2.2 Type 8.5-SW(AT)-1D/6.0-2G. Type 8.5-SW(AT)-1D/6.0-2G shall be a combination single drum, double gypsy, spanwire winch consisting of two independent winches mounted integrally on a single bedplate. Each unit shall be indepentently operated and have its own power unit and operating controls(see table I).

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3.3 Drums. Drums shall be designed for the following:

(a) Stowage capacity for 600 feet of 3/4 inch wire rope. The breaking strength of the wire rope is 48,600 pounds.
(b) Barrel diameter of 24 inches plus or minus 1/2 inch.
(c) Distance between flanges shall be 15 inches plus or minus 1/2 inch.

3.4 Wire rope construction. Wire rope for the winch shall be type 1, class 3, construction 6, independent wire rope core(IWRC), improved plow steel, uncoated, preformed, right regular lay in accordance with RR-W-410.

3.5 Gearing. The gear drives provided shall have a service factor of 1.25.

3.6 Sheaves. All sheaves shall be designed with a minimum sheave tread diameter to wire rope diameter ratio of 24 to 1.

3.7 Torque clutch. The drum shall be equipped with a torque (slip) clutch. This clutch shall be the driving member between the drum shaft and the drum. The tension (slip) value specified below is based on the wire being on the average layer of the drum. The clutch shall not use any external energy source and shall be self compensating for normal wear. The slip value of the clutch shall not be effected by environments specified for the winch or clutch inactivity (no slippage) for the design life of the winch.

3.7.1 The torque clutch for the winch shall slip at a maximum of 20,000 pounds and reset at a minimum of 15,000 pounds.

3.8 Controlled spooling and anti-slack devices. Each winch drum shall be provided with controlled spooling and an anti-slack device to insure proper spooling of wire on the drum. Proper spooling shall exist at all times the equipment is in operation as well as during start-up, after periods of disuse, without making adjustments. Proper spooling shall exist during periods of maximum winch deceleration and maximum acceleration between full speed haul-in and full speed payout for all winch loads from zero to rated load.

3.8.1 Controlled spooling. The controlled spooling device shall consist of a grooved drum, power operated level wind and a swiveling fairleader.

3.8.1.1 Grooved drum. The grooved drum shall be designed so that wire spools shall be provided on the drum in an evenly spaced, interrupted spiral pattern from flange to flange. The first and each succeeding layer of wire on the drum shall form a spooling pattern for the next layer that will exactly reproduce this spooling pattern on all layers. The spooling pattern of wire on the drum shall eliminate overriding (cross over) turns and undercutting of the wire rope and result in a minimum amount of scrubbing and chafing of the wire rope.

3.8.1.2 Level wind. A power operated level wind or similar device, automatically operated, shall be provided as a part of the spooling system.

3.8.1.3 Swiveling fairleader. A swiveling fairleader shall be provided as a part of the spooling system. The fairleader shall be designed so that it can be readily locked in any operating position. The swiveling fairleader shall be mounted on the winch foundation.

3.8.2 Anti-slack device. The anti-slack device shall be hydraulically driven and shall prevent slack wire from developing between the fairleader and the drum.

3.8.2.1 ON-OFF switch. A snap action ON-OFF switch shall be provided on the control console to permit the operator to activate or deactivate the anti-slack device.

3.8.2.2 Deactivating the anti-slack device. When the anti-slack device is deactivated the pumping unit and the hydraulic motor shall be disengaged or otherwise protected from overspeeding.

3.8.3 Wire rope life requirements. With the anti-slack device energized, no load on the winch and the winch operated at rated speed, the anti-slack device and controlled spooling system shall provide at least 7,500 cycles of operation without failure (see 3.8.7) of the wire rope. With the anti-slack device turned off, rated load on the winch and the winch operated at rated speed, the anti-slack device and controlled spooling system shall provide at least 15,000 cycles of operation without failure of the wire rope.

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3.8.4 Cycle. A cycle is defined as passage of a section of wire rope from the winch drum through the winch controlled spooling system and back onto the winch drum.

3.8.5 Cycle rate. Cycle rate when operating at winch rated speed and rated load is defined as one cycle every fifteen seconds.

3.8.6 Wire rope failure. Wire rope failure is defined as seven or more broken wires in one strand in one rope lay.

3.8.7 Controls. The winch controlled spooling system shall be provided with controls as necessary to permit initial spooling of wire rope on the winch and through the shipboard underway replenishment system. These controls shall be built into the master control panel (see 3.14.1.1).

3.8.8 Wire rope lead. The wire rope shall lead from the drum to the swiveling fairleader/anti-slack device via the level wind. The swiveling fairleader/anti-slack device shall be designed so that the wire can be led up at any angle up to 60 degrees each side of a vertical plane normal to the winch drum and simultaneously lead at any angle from 15 degrees down to 75 degrees above the horizontal.

3.9 Modes of operation. The winch shall be provided with speed and tension control modes of operation.

3.9.1 Speed control mode of operation. The rated line pull and rated line speed on the average layer for speed mode of operation shall be as follows:

(a) Rated line pull - 8500 pounds on the average layer of the drum.
(b) Rated line speed - 300 fpm.

3.9.2 Automatic tension mode of operation.

3.9.2.1 Line tension. Line tension shall be determined by a line sensing device.

3.9.2.2 Performance.

3.9.2.2.1 The automatic tension control of the winch shall be designed so that it will cause the winch to payout or haul-in to maintain required tension level.

3.9.2.2.2 The speed-tension values at maximum tension setting shall follow the curve formed by points A, B, C, and D (see figure 1). The speed at point A shall be 540 fpm (payout) and point D the speed shall be 540 fpm (haul-in). Curve CD shall be a constant power curve formed by the following equation:

Speed (fpm haul-in) times tension (pounds) equals 2065×10^6.

3.9.2.2.3 The curve from points A, B, and C (see figure 1) shall be a straight line and shall have a tension decrease of one pound for each fpm speed increase. The speed at point B shall be zero and the tension 8500 pounds.

3.9.2.2.4 The winch design shall permit the winch to be overhauled by an external force at speeds from 540 to 700 fpm. The tension value during this period may be any value required to protect winch components from damage including power generation, but shall not exceed 9200 pounds.

3.9.2.2.5 The winch design shall provide automatic recovery from the 540 to 700 fpm overhauling speed condition back to normal tension control range (540 fpm payout to 540 feet fpm haul-in).

3.9.2.2.6 The design of the automatic tension control shall be such that reference points A through D (see figure 1) could be modified either through adjustments or substitution of components.

3.9.2.2.7 Actual tension values shall be within $\pm$ 500 pounds of the values specified in 3.9.2.2.2 through 3.9.2.2.4.

3.9.2.2.8 The winch shall provide tension accuracy at all tension settings of $\pm$ 15 percent of the rated tension during acceleration and deceleration at rates up to 6.7 feet per second squared.

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3.9.2.2.9 The rate at which the tension level can be manually raised from minimum to maximum values shall be restricted to a minimum of three seconds.

3.9.2.2.10 The winch system shall be designed so that when tension mode is initially selected with slack wire present in the system, the winch shall haul-in at not more than 150 fpm line speed until a line tension of 1500 pounds is obtained. The winch shall then automatically switch from this slow haul-in mode of operation to normal speed-tension mode of operation.

3.10 Duty. Winch type 8.5-SW(AT)-1D and the drum unit of winch type 8.5-SW(AT)-1D/6.0-2G shall be designed to permit continuous operation for a period of six hours. A normal operating cycle will consist of 10 seconds at rated speed and rated load; 10 seconds at idle; 10 seconds at half-speed and rated load; and 10 seconds at idle. (See 3.16.2 for duty requirements for gypsy unit of winch type 8.5-SW(AT)-1D/6.0-2G.)

3.10.1 Nameplate ampere rating of the electric motor shall not be exceeded by more than twenty-five percent when the winch is being operated at rated load and rated speed (constant velocity), temperature rise shall not exceed nameplate permissible temperature rise when the winch is operated at the duty cycle and period specified in 3.10.

3.10.2 Thermal overload shall be indicated by energizing a master panel indicating light through a thermal heating device (see thermal requirements of MIL-W-17265) in either the motor or the controller. The indicator light shall be in accordance with the illumination requirements of MIL-W-17265. This device shall not de-energize the motor.

3.11 Hydraulic brake. When in speed control mode of operation the hydraulic brake shall automatically set upon loss of servo pressure or electric power. When in automatic tension mode of operation the hydraulic brake shall be designed to stay in the released position for at least fifteen minutes upon loss of servo pressure or electric power.

3.12 Relief valve. The high pressure relief valve (hoist side) for the hydraulic system shall automatically vent to a lower pressure valve to limit the maximum load that can be imposed on the wire rope, upon loss of electric power to the winch.

3.12.1 The lower pressure for the relief valve shall be adjustable and shall be set at a value that will permit the winch to be overhauled with a rope pull of 3500 to 4000 pounds (wire rope on the average layer) when operating in tension mode of operation. This requirement is based on the hydraulic brake being released (see 3.11).

3.13 Hydraulic motors. The hydraulic motor may be of the variable delivery type. If a variable delivery type of hydraulic motor is provided a position indicator shall be provided to indicate the setting of the motor. In the event of loss of power to the system the hydraulic motor, if variable delivery type, shall go to minimum stroke.

3.14 Methods of operation. Master speed control and winch speed control units conforming to the requirements of MIL-W-17265 shall be provided. In addition a master tension control unit conforming to 3.14.1 shall be provided.

3.14.1 Master tension control. A master tension control shall be added to the operations selector switch (see MIL-W-17265 for requirements for master control panel).

3.14.1.1 Associated controls required to perform the functions of 3.8.7 and 3.9.2 shall also be added to the master control panel.

3.14.1.2 An indicator light shall be added to the panel for each method of operation. The lights and the dimmer control shall be in accordance with the requirements of MIL-W-17265 for illumination.

3.14.2 Replenishment at sea control stations. Replenishment at sea control stations shall be of an integrated console design in lieu of miscellaneous arrangement of control and indicating lights.

3.14.2.1 A master control panel shall consist of all controls and indicating lights for the spanwire winches.

3.15 Size and weight. The winch shall be as compact and light as is consistent with adequate strength and rigidity. Weight and size shall not exceed the following:

(a) Type 8.5-SW(AT)-1D:
    (1) Maximum weight - 12,000 pounds.

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(2) Maximum width (across drum) with gypsy head installed - 90 inches.
(3) Maximum length (perpendicular to drum) - 104 inches.
(4) Maximum height - 105 inches.

(b) Type 8.5-SW(AT)-1D/6.0-2G:
(1) Maximum weight - 13,500 pounds.
(2) Maximum width (across drum) with gypsy heads installed - 96 inches.
(3) Maximum length (perpendicular to drum) - 140 inches.
(4) Maximum height - 105 inches.

3.16 Gypsy winch. The gypsy winch unit of winch type 8.5-SW(AT)-1D/6.0-2G shall be in accordance with MIL-W-15802, except as specified herein. Requirements of MIL-W-15802 that conflict with those of MIL-W-17265, the requirements of MIL-W-17265 shall govern.

3.16.1 Gypsy heads. Gypsy heads for the gypsy unit of winch type 8.5-SW(AT)-1D/6.0-2G shall be proportioned for five inch circumference manila rope, maximum breaking strength of 22,500 pounds.

3.16.2 Gypsy winch electric motor. The gypsy winch electric motor for winch type 8.5-SW(AT)-1D/6.0-2G shall be in accordance with MIL-M-17060 and the following:

Classification	Service A
Ambient temperature	40 degrees C
Voltage	440 ac
Phase	3
Hertz	60
Duty	Full load, 1/2 hour; 1/2 load, continuous
Enclosure	Watertight
Speed	Two speed (full and half) constant torque
Type	Squirrel cage induction
Design	D
Horsepower	As required

3.16.3 Gypsy winch motor controller. The gypsy winch electric motor controller and master switch for winch type 8.5-SW(AT)-1D/6.0-2G shall be in accordance with MIL-C-2212 or MIL-C-23742, as applicable and the following:

Operation	Magnetic
Voltage	440 vac, 3 phase, 60 hertz
Type	Across-the-line
Performance	Non-automatic
Function	Motor starting, reversing and speed selection
Protection	Low voltage release and thermal overload
Enclosure	Dripproof protected
Duty	Continuous
Emergency run feature	Remote emergency run pushbutton shall be provided with a watertight housing
Ambient temperature	50 degrees C
Master switch	Remote drum type, lever operated spring return to OFF, watertight enclosure with power ON indicating light

3.16.4 Gypsy winch electric brake. The gypsy winch electric brake for winch type 8.5-SW(AT)-1D/6.0-2G shall be in accordance with MIL-B-16392 and the following:

Voltage	440 ac
Duty	Continuous
Ambient temperature	40 degrees C
Enclosure	Watertight
Mounting	As required

3.16.5 Gypsy winch rated line pull and line speed. The rated line pull and rated line speed of the winch shall be as follows:

(a) Rated line pull - 6000 pounds.
(b) Rated line speed - 120 and 60 fpm

4. QUALITY ASSURANCE PROVISIONS

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4.1 The quality assurance provisions shall be as specified in MIL-W-17265, except as specified herein.

4.2 Quality conformance inspection.

4.2.1 Examination. In addition to the examinations listed in table II of MIL-W-17265, the examinations listed in table I herein shall be conducted.

Table I - Examination.

Examination	Requirement paragraph
Verify winch is correct type. Either type 8.5-SW(AT)-1D or 8.5-SW(AT)-1D/6.0-2G.	3.2.1 and 3.2.2
Drum will stow 600 feet of 3/4-inch wire rope	3.3(a)
Barrel diameter of drum is 24-inches plus or minus 1/2-inch.	3.3(b)
Distance between flanges of drum is 15-inches plus or minus 1/2-inch.	3.3(c)
All sheaves are designed with a minimum sheave tread diameter to wire rope diameter of 24 to 1.	3.6
Torque clutch is driving member between the drum shaft and the drum.	3.7
Wire rope can be led from the winch in the following manner:	3.8.8
(a) Led up to 60 degrees each way from a vertical plane normal to the drum.	
(b) Led at any angle from 15 degrees below to 75 degrees above horizontal.	
The winch provides both tension and speed modes of operation.	3.9

4.2.2 Tests. In addition to the tests specified in MIL-W-17265, the tests specified in 4.3.1 through 4.3.5.4 shall be conducted.

4.3 Test methods.

4.3.1 Controlled spooling system and wire rope life test. One winch of each type offered for delivery on one contract or order shall be tested to show conformance with all requirements specified for the controlled spooling system and for wire rope life (see 3.8 through 3.8.7).

4.3.2 Torque clutch tests.

4.3.2.1 Each drum torque clutch shall be tested by pulling a 40 foot length of wire off the drum at a speed of 300 fpm. The maximum tension measured during this test shall not exceed 20,000 pounds. (The use of a special locking device on the winch to hold the drum shaft stationary during this test is acceptable).

4.3.2.2 The torque clutch of one winch of each type offered for delivery on the contract or order shall be tested by repeating the test specified in 4.3.2.1 two times, waiting approximately 15 seconds between runs. Repeat this test seven additional times allowing the clutch to cool as necessary between runs. The maximum tension measured during each run shall be recorded. The maximum tension measured during this test shall not exceed 20,000 pounds.

4.3.3 Vented relief valve test. Each winch shall be tested to demonstrate that the vented relief valve has been properly set.

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4.3.4 Tension test. One winch of each type offered for delivery on the contract or order shall be tested to show conformance with the requirements for the tension mode of operation (see 3.9.2 through 3.9.2.2.10).

4.3.5 Gypsy winch test (type 8.5-SW(AT)-1D(6.0-2G). The gypsy winch unit of winchtype 8.5-SW(AT)-1D(6.0-2G) shall be tested in accordance with 4.3.5.1 through 4.3.5.4.

4.3.5.1 No load. Each winch shall be operated under no load for a period of 1/2 hour in each direction of rotation. No abnormal heating (150°F) of bearings or operating ports shall occur.

4.3.5.2 Rated load. Each winch shall be operated, hoisting and lowering rated load at rated speed through a distance of at least 20 feet in each direction for 30 minutes. The load shall be stopped and suspended for approximately 45 seconds at the end of each cycle.

4.3.5.3 Overload. Each winch shall demonstrate its ability to hoist and lower 150 percent rated load at no specified speed through a distance of at least 20 feet in each direction for five complete cycles. The load shall be stopped and held suspended for a period of approximately 45 seconds at the end of each cycle.

4.3.5.4 Electric brake. The electric brake of each winch shall be subjected to a static pull equal to 200 percent of the rated live pull of the winch without slippage. Each electric brake shall also demonstrate its ability to stop and hold a load equal to 150 percent of the rated load of the winch when the load is lowered at rated speed under simulated power conditions. Stopping distance under this test shall not exceed six feet of wire rope travel.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-W-17265.

6. NOTES

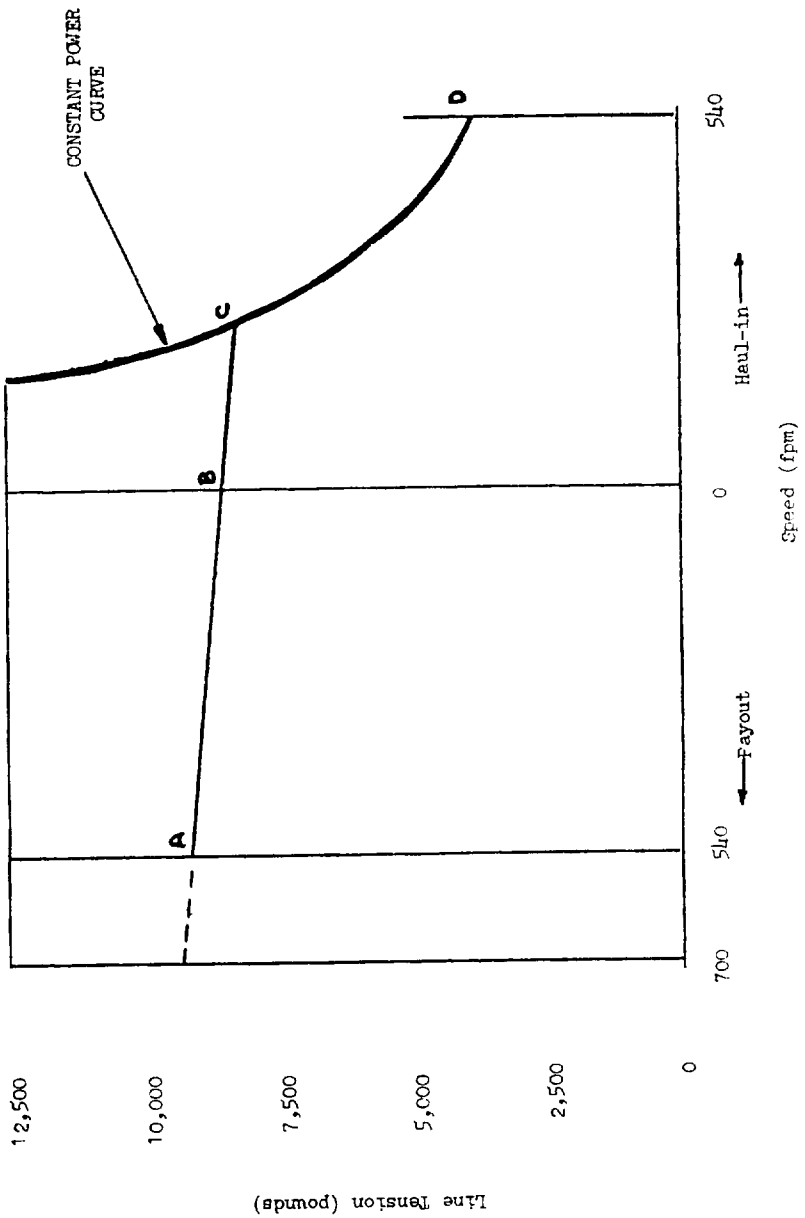
6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).

6.2 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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Figure 1 - Speed-tension curve.

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